

I need a responsive company website with Home, About and Con

Status: Approved by reviewer

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Domains: frontend

1. Executive summary

I need a responsive company website with Home, About and Contact pages. Frontend only.

Project type: Mock project - medium complexity. Mock estimate.

At a glance: 2.5 FTE team, about 47 person-days of effort, about 7 weeks to deliver.

Reviewer: Solution matches the scope; security gap closed.

2. Solution architecture

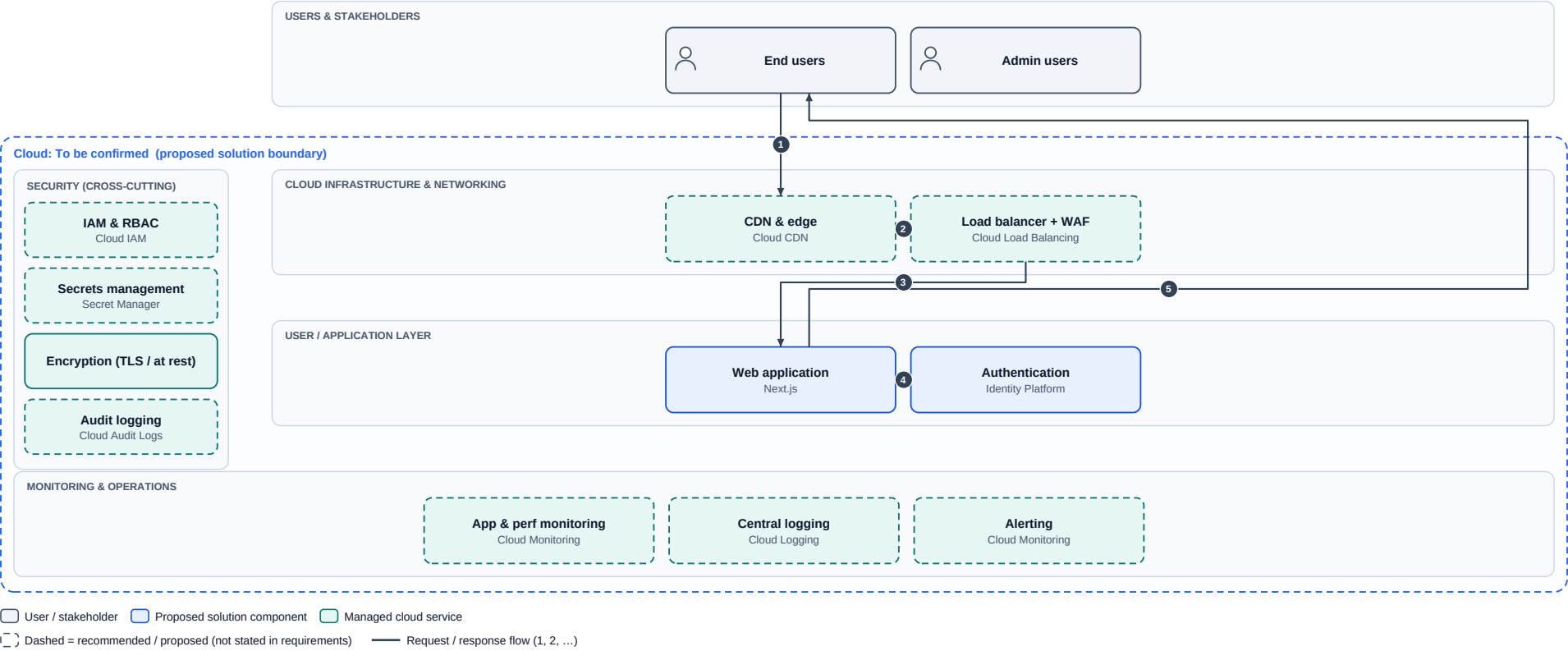
Mock architecture: users reach a web application through a managed edge; services, data and (where needed) AI components run inside one cloud boundary.

2.1 High-level solution architecture diagram

High-level solution architecture

I need a responsive company website with Home, About and Con

High-level solution architecture · cloud: To be confirmed



Cloud: To be confirmed. Boxes are grouped by layer and coloured by type (client system, proposed component, third-party, managed cloud service, AI, data store). Dashed boxes are recommended additions that the requirement did not state.

2.2 End-to-end data flow

Request / response

- 1. End users → CDN & edge: HTTPS request
- 2. CDN & edge → Load balancer + WAF: Route + filter
- 3. Load balancer + WAF → Web application: Serve app
- 4. Web application → Authentication: Sign in
- 5. Web application → End users: Rendered page

2.3 Major components

Layer	Component	Technology	Type	Status	Purpose
Users & stakeholders	End users	-	actor	required	Mock: end users
Users & stakeholders	Admin users	-	actor	required	Mock: admin users
User / application layer	Web application	Next.js	proposed	required	Mock: web application
User / application layer	Authentication	Identity Platform	proposed	required	Mock: authentication
Cloud infrastructure & networking	CDN & edge	Cloud CDN	cloud service	recommended	Mock: cdn & edge
Cloud infrastructure & networking	Load balancer + WAF	Cloud Load Balancing	cloud service	recommended	Mock: load balancer + waf
Security	IAM & RBAC	Cloud IAM	cloud service	recommended	Mock: iam & rbac
Security	Secrets management	Secret Manager	cloud service	recommended	Mock: secrets management
Security	Encryption (TLS / at rest)	-	cloud service	required	Mock: encryption (tls / at rest)
Security	Audit logging	Cloud Audit Logs	cloud service	recommended	Mock: audit logging
Monitoring & operations	App & perf monitoring	Cloud Monitoring	cloud service	recommended	Mock: app & perf monitoring
Monitoring & operations	Central logging	Cloud Logging	cloud service	recommended	Mock: central logging
Monitoring & operations	Alerting	Cloud Monitoring	cloud service	recommended	Mock: alerting

2.4 Requirement traceability

Requirement	Delivered by
As described in the requirement	IAM & RBAC, Secrets management
Responsive	Web application, Authentication
Secure	IAM & RBAC, Secrets management
Maintainable	Web application, Authentication

2.5 Key assumptions

- Mock: single region deployment
- Mock: corporate SSO is available

2.6 Recommended technology stack

Layer	Technology	Purpose	Status
User / application layer	Next.js	Mock: web application	required
User / application layer	Identity Platform	Mock: authentication	required
Cloud infrastructure & networking	Cloud CDN	Mock: cdn & edge	recommended
Cloud infrastructure & networking	Cloud Load Balancing	Mock: load balancer + waf	recommended
Security	Cloud IAM	Mock: iam & rbac	recommended
Security	Secret Manager	Mock: secrets management	recommended
Security	Cloud Audit Logs	Mock: audit logging	recommended
Monitoring & operations	Cloud Monitoring	Mock: app & perf monitoring	recommended
Monitoring & operations	Cloud Logging	Mock: central logging	recommended
Monitoring & operations	Cloud Monitoring	Mock: alerting	recommended

2.7 Security considerations

- Mock: TLS everywhere, least-privilege IAM, secrets in a vault

2.8 Scalability and future enhancements

Scalability

- Mock: stateless services scale horizontally

Future enhancements

- Mock: add analytics warehouse once usage grows

3. Solution design

Recommended architecture

Style: Mock architecture

Mock overview of the reviewed design.

- Presentation: mock
- Services: mock

Agents used

- **frontend:** needed for frontend
- **uiux:** needed for frontend
- **security:** needed for frontend
- **performance:** needed for frontend

Components

Component	Responsibility	From
Frontend module	Handles frontend concerns	frontend
Uiux module	Handles uiux concerns	uiux
Security module	Handles security concerns	security
Performance module	Handles performance concerns	performance

Recommended technology stack

Category	Choice	Rationale	Alternatives	From
frontend_framework	Next.js	SSR/SSG for fast, SEO-friendly pages	Astro, Nuxt	frontend
styling_ui	Tailwind CSS	Fast responsive layouts	CSS Modules	frontend
styling_ui	shadcn/ui	Accessible, unstyled primitives	Material UI	uiux
auth	Identity Platform	Managed authentication service	Auth0, Keycloak	security
testing_quality	Lighthouse	Core Web Vitals checks in CI	k6	performance

4. Resources and estimate

Required human resources

Role	FTE	Seniority	Responsibilities	Phases
Frontend developer	1	mid	Mock: frontend developer work	Build
UI/UX designer	0.5	mid	Mock: ui/ux designer work	Build
Security engineer	0.25	senior	Mock: security engineer work	Build
QA engineer	0.5	mid	Mock: testing	Testing
Project manager	0.25	senior	Mock: delivery	Discovery, Build, Testing

Team size: 2.5 FTE across 5 roles.

Required AI and technical resources

Type	Resource	Purpose	Sizing
hosting	Vercel	Mock hosting	Pro plan
testing	Playwright	Mock E2E tests	CI runs

No AI models or services are needed for this solution.

Estimated development effort

Phase	Roles	Low	Likely	High
Discovery	Frontend developer, Project manager	3	5	7
Build	Frontend developer, UI/UX designer, Security engineer	26.25	35	43.75
Testing	QA engineer, Frontend developer	5	7	10
Total (person-days)		34.25	47	60.75

About **2.4 person-months** likely (20 working days per month).

Estimated timeline

Phase	Weeks	Duration	Depends on	Schedule
Discovery	1	1 wk	-	
Build	2–5	4 wk	Discovery	
Testing	6–7	2 wk	Build	

Total: about 7 weeks with the team above (range 5.1–9 weeks, following the effort range).

What could change the estimate:

- Mock risk to estimate

5. Specialist findings

frontend

Mock frontend design for the requirement.

- Mock design note from the frontend agent.

uiux

Mock uiux design for the requirement.

- Mock design note from the uiux agent.

security

Mock security design for the requirement.

- Mock design note from the security agent.
- Added MFA for admin users and rate limiting on all authentication endpoints.

performance

Mock performance design for the requirement.

- Mock design note from the performance agent.

6. Alternatives

Listed per technology choice in section 3.

7. Risks

Risk	Severity	Likelihood	Mitigation	From
Mock frontend risk	low	low	Review during pilot	frontend
Mock uiux risk	low	low	Review during pilot	uiux
Mock security risk	low	low	Review during pilot	security
Mock performance risk	low	low	Review during pilot	performance

8. Quality score

Overall: 88.0/100

Dimension	Weight %	Score
requirement_fit	25.0	88
architecture	20.0	88
accessibility	15.0	88
security	15.0	88
performance	15.0	88
maintainability	10.0	88

9. Assumptions

- Mock analysis: replace with a real model for meaningful output
- frontend: mock assumption
- uiux: mock assumption
- security: mock assumption
- performance: mock assumption
- Mock estimate: replace with a real model for meaningful numbers

10. Review history

Round	Decision	Score	Findings (severity)
1	REWORK	74.0	high: No MFA or rate limiting on authentication endpoints
2	APPROVED	88.0	low: Add a component diagram